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Course Code:CSA1709
Course Name:Artificial Intelligence

Exp :3 Write the Python Program for Water Jug Problem

Aim:

To Write the Python Program for Water Jug Problem

Algorithm:

Step1: Start with two empty jugs of different capacities, and a target quantity of water to measure.

Step2: Fill one jug to its maximum capacity.

Step3: Pour the water from the full jug into the other jug until it is full or the first jug is empty.

Step4: If the second jug is full, pour out the water to empty it.

Step5: Repeat steps 2-4, filling and pouring water between the two jugs, until the desired quantity of water is measured or all possible combinations are tried.

Program:

```
from collections import deque
```

```
def water_jug_solver(capacity1, capacity2, target):
```

```
    visited = set()
```

```
    queue = deque([(0, 0), []])
```

```
    while queue:
```

```
        (jug1, jug2), path = queue.popleft()
```

```
        if (jug1, jug2) in visited:
```

```
            continue
```

```
        visited.add((jug1, jug2))
```

```
current_path = path + [(jug1, jug2)]
```

```
if jug1 == target or jug2 == target:
```

```
    return current_path
```

```
possible_states = [
```

```
    (capacity1, jug2),
```

```
    (jug1, capacity2),
```

```
    (0, jug2),
```

```
    (jug1, 0),
```

```
    (jug1 - min(jug1, capacity2 - jug2), jug2 + min(jug1, capacity2 - jug2)),
```

```
    (jug1 + min(jug2, capacity1 - jug1), jug2 - min(jug2, capacity1 - jug1))
```

```
]
```

```
for state in possible_states:
```

```
    if state not in visited:
```

```
        queue.append((state, current_path))
```

```
return None
```

```
if __name__ == '__main__':
```

```
    jug1_capacity = 4
```

```
    jug2_capacity = 3
```

```
    target_amount = 2
```

```
solution = water_jug_solver(jug1_capacity, jug2_capacity, target_amount)
```

```
if solution:
```

```
    print("Steps to solve the water jug problem:")
```

```
    for step_num, step in enumerate(solution):
```

```
        print("Step", step_num, ": Jug 1 =", step[0], ", Jug 2 =", step[1])
```

```
else:
```

```
print("No solution found.")
```

OUTPUT:

```
Steps to solve the water jug problem:  
Step 0 : Jug 1 = 0 , Jug 2 = 0  
Step 1 : Jug 1 = 0 , Jug 2 = 3  
Step 2 : Jug 1 = 3 , Jug 2 = 0  
Step 3 : Jug 1 = 3 , Jug 2 = 3  
Step 4 : Jug 1 = 4 , Jug 2 = 2
```

Result: Thus, the program was executed and found that the output was found to be correct.